

# Simulation study: mixture of experts model

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Data simulation:

- Sample size  $n = 200$
- Dimension of the Wishart distribution  $p = 2$
- Number of latent components  $K = 3$
- Probabilities of subpopulations  $\boldsymbol{\pi} = (0.35, 0.40, 0.25)$
- Degrees of freedom  $\boldsymbol{\nu} = (8, 12, 3)$
- Scale matrices of the Wishart distribution  $\Sigma_1, \Sigma_2, \Sigma_3$
- Data  $S_i \sim \pi_1 \text{Wishart}(\nu_1, \Sigma_1) + \pi_2 \text{Wishart}(\nu_2, \Sigma_2) + \pi_3 \text{Wishart}(\nu_3, \Sigma_3)$

Modeling:

- Bayesian mixture of experts of two Wishart distributions

```
rm(list = ls())

library(moewishart)

set.seed(123)
n <- 200 # number of subjects
p <- 2

dat <- simData(n, p,
  pis = c(0.35, 0.40, 0.25),
  nus = c(8, 12, 3),
  Sigma = NULL # default Sigmas pre-defined in function 'simData()'
)
S_list <- dat$S
Sigma_list <- dat$Sigma_list

my_K_intial <- 3

# run a moewishart model
start_time <- Sys.time() # this has to be before 'set.seed()', otherwise non-reproducible
set.seed(123)
fit <- moewishart(S_list,
  K = my_K_intial, init_pi = c(0.45, 0.40, 0.15), mh_sigma = 1,
  niter = 10000, burnin = 1000, thin = 1, verbose = TRUE
)

Iter    1 | LL=-828.3 | 59.8 iter/sec | acc_rate_nu=[0.000 0.000]
Iter  500 | LL=-780.3 | 861.8 iter/sec | acc_rate_nu=[0.042 0.146]
Iter 1000 | LL=-779.2 | 881.1 iter/sec | acc_rate_nu=[0.044 0.133]
Iter 1500 | LL=-778.9 | 888.4 iter/sec | acc_rate_nu=[0.038 0.131]
Iter 2000 | LL=-777.6 | 900.0 iter/sec | acc_rate_nu=[0.039 0.130]
```

```

Iter 2500 | LL=-780.0 | 902.1 iter/sec | acc_rate_nu=[0.036 0.127]
Iter 3000 | LL=-777.3 | 898.9 iter/sec | acc_rate_nu=[0.036 0.132]
Iter 3500 | LL=-780.0 | 897.5 iter/sec | acc_rate_nu=[0.037 0.126]
Iter 4000 | LL=-779.1 | 897.7 iter/sec | acc_rate_nu=[0.035 0.127]
Iter 4500 | LL=-777.1 | 898.2 iter/sec | acc_rate_nu=[0.036 0.126]
Iter 5000 | LL=-776.9 | 898.3 iter/sec | acc_rate_nu=[0.037 0.120]
Iter 5500 | LL=-775.5 | 891.3 iter/sec | acc_rate_nu=[0.036 0.115]
Iter 6000 | LL=-773.2 | 892.5 iter/sec | acc_rate_nu=[0.036 0.110]
Iter 6500 | LL=-770.2 | 894.9 iter/sec | acc_rate_nu=[0.036 0.104]
Iter 7000 | LL=-777.0 | 896.1 iter/sec | acc_rate_nu=[0.035 0.101]
Iter 7500 | LL=-775.0 | 896.3 iter/sec | acc_rate_nu=[0.035 0.098]
Iter 8000 | LL=-769.6 | 896.2 iter/sec | acc_rate_nu=[0.035 0.096]
Iter 8500 | LL=-773.2 | 896.5 iter/sec | acc_rate_nu=[0.034 0.093]
Iter 9000 | LL=-768.1 | 898.6 iter/sec | acc_rate_nu=[0.035 0.091]
Iter 9500 | LL=-778.8 | 900.6 iter/sec | acc_rate_nu=[0.035 0.090]
Iter 10000 | LL=-775.1 | 899.8 iter/sec | acc_rate_nu=[0.036 0.088]

```

```

end_time <- Sys.time()
elapsed_time <- end_time - start_time
elapsed_time

```

Time difference of 11.1237 secs

```

# posterior summaries
colMeans(fit$pi) # estimated pi

```

```
[1] 0.2451644 0.3977296 0.3571060
```

```
colMeans(dat$pi) # true pi
```

```
[1] 0.35 0.40 0.25
```

```
apply(fit$nu, 2, mean) # estimated nu
```

```
[1] 6.064036 13.235072 6.941049
```

```
dat$nu # true nu
```

```
[1] 8 12 3
```

```

# Sum all arrays in the list and divide by the number of saved iterations
sigma_posterior_mean <- Reduce("+", fit$Sigma) / length(fit$Sigma)
# View the mean covariance matrix for Cluster 1
print(sigma_posterior_mean[, , 1])

```

```

      [,1]      [,2]
[1,] 2.2247971 -0.1548921
[2,] -0.1548921 2.1203926

```

```
print(sigma_posterior_mean[, , 2])
```

```

      [,1]      [,2]
[1,] 1.8853488 0.4230219
[2,] 0.4230219 1.3683697

```

```
print(sigma_posterior_mean[, , 3])
```

```

      [,1]      [,2]
[1,] 0.7274027 0.3019724
[2,] 0.3019724 0.9266380

```

```
as.data.frame(Sigma_list)
```

```
      X1  X2 X1.1 X2.1 X1.2 X2.2  
1 0.5 0.2  2.0  0.6  4.0  0.2  
2 0.2 0.7  0.6  1.5  0.2  3.0
```

```
(a1 <- apply(sigma_posterior_mean, MARGIN = 3, function(aa) min(sapply(Sigma_list, function(x) sum((aa
```

  
[1] 1.57514502 0.09311395 0.12387348

## Check identifiability issue

```
# re-run the model with wrong initial weights to check identifiability issue  
set.seed(1234)  
fit2 <- moewishart(S_list,  
  K = my_K_intial, init_pi = c(0.15, 0.40, 0.45), mh_sigma = 1,  
  niter = 10000, burnin = 1000, thin = 1, verbose = TRUE  
)
```

```
Iter    1 | LL=-828.3 | 848.9 iter/sec | acc_rate_nu=[0.000 1.000]  
Iter   500 | LL=-779.6 | 921.8 iter/sec | acc_rate_nu=[0.052 0.142]  
Iter  1000 | LL=-779.8 | 916.2 iter/sec | acc_rate_nu=[0.038 0.135]  
Iter  1500 | LL=-775.9 | 913.3 iter/sec | acc_rate_nu=[0.043 0.111]  
Iter  2000 | LL=-770.3 | 911.6 iter/sec | acc_rate_nu=[0.043 0.103]  
Iter  2500 | LL=-770.0 | 911.2 iter/sec | acc_rate_nu=[0.043 0.097]  
Iter  3000 | LL=-769.3 | 909.1 iter/sec | acc_rate_nu=[0.043 0.094]  
Iter  3500 | LL=-769.3 | 910.8 iter/sec | acc_rate_nu=[0.041 0.087]  
Iter  4000 | LL=-767.2 | 914.9 iter/sec | acc_rate_nu=[0.042 0.086]  
Iter  4500 | LL=-773.1 | 917.8 iter/sec | acc_rate_nu=[0.041 0.084]  
Iter  5000 | LL=-774.6 | 917.6 iter/sec | acc_rate_nu=[0.039 0.084]  
Iter  5500 | LL=-775.7 | 920.0 iter/sec | acc_rate_nu=[0.039 0.083]  
Iter  6000 | LL=-773.8 | 922.2 iter/sec | acc_rate_nu=[0.037 0.082]  
Iter  6500 | LL=-769.9 | 921.1 iter/sec | acc_rate_nu=[0.038 0.082]  
Iter  7000 | LL=-770.7 | 920.8 iter/sec | acc_rate_nu=[0.038 0.080]  
Iter  7500 | LL=-773.9 | 921.5 iter/sec | acc_rate_nu=[0.038 0.079]  
Iter  8000 | LL=-772.4 | 920.1 iter/sec | acc_rate_nu=[0.038 0.078]  
Iter  8500 | LL=-774.6 | 919.4 iter/sec | acc_rate_nu=[0.038 0.076]  
Iter  9000 | LL=-772.8 | 918.4 iter/sec | acc_rate_nu=[0.037 0.076]  
Iter  9500 | LL=-773.2 | 918.9 iter/sec | acc_rate_nu=[0.038 0.074]  
Iter 10000 | LL=-770.2 | 918.7 iter/sec | acc_rate_nu=[0.038 0.074]
```

```
colMeans(fit2$pi) # estimated pi
```

```
[1] 0.2862457 0.3977179 0.3160364
```

```
colMeans(dat$pi) # true pi
```

```
[1] 0.35 0.40 0.25
```

```
apply(fit2$nu, 2, mean) # estimated nu
```

```
[1] 7.888912 12.375095 3.417019
```

```
dat$nu # true nu
```

```
[1] 8 12 3
```

```
# Sum all arrays in the list and divide by the number of saved iterations
sigma_posterior_mean2 <- Reduce("+", fit2$Sigma) / length(fit2$Sigma)
# View the mean covariance matrix for Cluster 1
print(sigma_posterior_mean2[, , 1])
```

```
      [,1]      [,2]
[1,] 0.4964003 0.2508158
[2,] 0.2508158 0.7331403
```

```
print(sigma_posterior_mean2[, , 2])
```

```
      [,1]      [,2]
[1,] 1.9740528 0.4545459
[2,] 0.4545459 1.4349301
```

```
print(sigma_posterior_mean2[, , 3])
```

```
      [,1]      [,2]
[1,] 2.8803092 0.1523065
[2,] 0.1523065 2.5718397
```

```
as.data.frame(Sigma_list)
```

```
      X1  X2 X1.1 X2.1 X1.2 X2.2
1 0.5 0.2  2.0  0.6  4.0  0.2
2 0.2 0.7  0.6  1.5  0.2  3.0
```

## EM algorithm

```
# run the model with EM algorithm
set.seed(1234)
fit3 <- moewishart(S_list, method = "em",
  K = my_K_intial, init_pi = c(0.45, 0.40, 0.15), mh_sigma = 1,
  niter = 10000, burnin = 1000, thin = 1, verbose = TRUE
)
```

```
Iter 10 | Loglik: -1841.9922 | Nu: 5.54, 4.87, 9.65
Iter 20 | Loglik: -1827.1686 | Nu: 6.13, 3.35, 11.63
Iter 30 | Loglik: -1826.0981 | Nu: 6.84, 3.16, 11.89
Iter 40 | Loglik: -1825.9483 | Nu: 7.15, 3.17, 11.97
Iter 50 | Loglik: -1825.9076 | Nu: 7.31, 3.19, 12.05
Iter 60 | Loglik: -1825.8940 | Nu: 7.38, 3.2, 12.11
Iter 70 | Loglik: -1825.8886 | Nu: 7.43, 3.21, 12.16
Iter 80 | Loglik: -1825.8862 | Nu: 7.45, 3.22, 12.2
Iter 90 | Loglik: -1825.8850 | Nu: 7.46, 3.22, 12.23
Iter 100 | Loglik: -1825.8843 | Nu: 7.47, 3.22, 12.26
Iter 110 | Loglik: -1825.8840 | Nu: 7.47, 3.23, 12.28
Iter 120 | Loglik: -1825.8838 | Nu: 7.48, 3.23, 12.29
Iter 130 | Loglik: -1825.8837 | Nu: 7.48, 3.23, 12.3
Iter 140 | Loglik: -1825.8836 | Nu: 7.48, 3.23, 12.31
Iter 150 | Loglik: -1825.8836 | Nu: 7.48, 3.23, 12.31
Iter 160 | Loglik: -1825.8836 | Nu: 7.48, 3.23, 12.32
Converged at iteration 166
```

```
# EM estimates
fit3$pi
```

```
[1] 0.3021154 0.2993243 0.3985603
```

```
colMeans(dat$pi) # true pi
```

```
[1] 0.35 0.40 0.25
```

```
fit3$nu
```

```
[1] 7.480525 3.229805 12.319807
```

```
dat$nu # true nu
```

```
[1] 8 12 3
```